

Your Perception Monitor May Be Measuring Nothing: A Null-Detector Diagnostic and a Cross-Session Validity Failure in Conformal Monitoring for Autonomous Navigation

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Abstract

Autonomous navigation stacks are built to trust their perception layer, and increasingly to monitor it: conformal prediction calibrates a nonconformity score on nominal data, an anytime-valid sequential test tracks its miscoverage on the live stream, and an alarm tells the planner when perception has stopped being trustworthy. We set out to reduce the detection delay of such a monitor for camera/LiDAR 3D detection under real adverse weather. We did not achieve that. What we report instead are two validated findings about how monitors of this kind fail silently, and an honest account of a weather effect we cannot resolve.

The first finding is that a conformal monitor can produce entirely plausible calibrated quantiles and alarm times while measuring nothing about the detector at all. If the detector’s regression head degenerates, the nonconformity score $\|B_{\text{pred}} - B_{\text{target}}\|_1$ collapses to $\|B_{\text{target}}\|_1$ — a function of the ground-truth labels alone — and the monitor reports the dataset’s annotation statistics as perception degradation. We introduce a *null-detector diagnostic* that exposes this: rerun the pipeline with the prediction head’s output replaced by zeros. It needs no additional data, training or annotation. Applied to our own earlier results it reproduced them *bit-identically* — nominal-to-degraded jump 0.13904668284371138 from both the real and the zeroed detector, agreeing to 17 significant figures. A monitor with no detector performed identically to one

with a detector, and on one dataset the null monitor was *better behaved* than the real one. We recommend this control as standard practice for conformal monitoring of perception, and we are explicit about its scope: it is a null for scores built on regression residuals, and a score reading a different head needs its own null.

The second finding is that split conformal’s exchangeability assumption fails across driving sessions. On the Canadian Adverse Driving Conditions dataset, per-drive miscoverage under one pooled quantile ranges from 0.083 to 0.623 against a target of 0.20 — a $7.5\times$ spread across drives that are all nominal, all clear-road, several sharing a collection date — with between-session variation as large as the weather effect being sought. The guarantee remains formally correct and becomes operationally useless: the monitor detects which session the vehicle is on. For navigation this is the worst available failure mode, because the alarm fires on route identity rather than on perception loss.

On the weather effect itself we report an *underpowered null*, and distinguish that carefully from an absence. On drive-disjoint held-out data with two-level clustered resampling, the effect is $+0.064$ with a 95% interval of $[-0.061, +0.193]$: consistent with a real effect of the size we previously claimed, and equally consistent with none. Nine evaluation sessions cannot resolve it; the minimum detectable effect is roughly twice the estimate. We also show *why* an earlier version of this measurement appeared significant, by reporting the interval with and without the degraded arm resampled — a decomposition that lets a reader distinguish "underpowered" from "absent" rather than guess, and which we suggest reporting alongside any clustered effect estimate of this kind.

Keywords: Conformal prediction, Perception monitoring, Autonomous navigation, Negative controls, Exchangeability, Anytime-valid sequential testing, Adverse weather, Reproducibility

1. Introduction

An autonomous navigation stack is architected to trust its perception layer. Path planning, risk assessment and minimum-risk manoeuvre selection all consume detections as though they were reliable, and the safety case for the vehicle rests on that reliability holding under the conditions the vehicle actually meets. Because it will not always hold, a monitoring layer is attractive: something that watches perception during operation and tells the planner when its inputs have stopped being trustworthy, so the stack can slow down, hand back, or pull over.

Conformal prediction offers a principled way to build one. Calibrate a nonconformity score on nominal data, and the fraction of new predictions falling outside the conformal region is controlled at a chosen level without distributional assumptions. Pair it with anytime-valid sequential testing — a betting process whose wealth is a nonnegative supermartingale under the nominal hypothesis — and the monitor can be interrogated continuously without a multiple-testing penalty, which is exactly what a vehicle needs from a system asked “has anything changed yet?” at every frame.

We set out to reduce the detection delay of such a monitor for camera/LiDAR 3D detection under real adverse weather. We did not achieve that, and this paper is about what we found instead. Two findings concern how monitors of this design fail *silently* — continuing to emit calibrated quantiles and plausible alarm times while not doing their job — and both matter more for navigation than a delay improvement would have.

A monitor can measure nothing and look healthy.. The nonconformity score used throughout this literature is a residual between predicted and ground-truth boxes. If the detector’s regression head degenerates — as ours did, under a placeholder training loss whose optimum is a constant — that residual becomes $\|B_{\text{target}}\|_1$, a function of the labels alone. Every downstream component then behaves exactly as specified: calibration returns a well-defined quantile, because it is the quantile of *some* exchangeable score distribution; the betting process accumulates wealth, because the miscoverage rate still moves; the alarm fires

at a plausible time, because annotation statistics genuinely differ between a dataset’s nominal and adverse splits. The monitor reports dataset structure as perception degradation, and nothing in its output reveals this.

We introduce a *null-detector diagnostic* that does reveal it: rerun the whole
35 pipeline, calibration included, with the prediction head’s output replaced by zeros. It costs one evaluation pass and needs no extra data, training or annotation. Applied to our own earlier results it reproduced them bit-identically — a nominal-to-degraded jump of 0.13904668284371138 from both the real and the zeroed detector, agreeing to 17 significant figures (Section 4.2). A monitor with
40 no detector performed identically to one with a detector. On CADC the null monitor was in fact *better behaved*, holding a zero false-alarm rate where the real one alarmed on everything — an inversion we explain in Section 5.1 and which is precisely why the failure is invisible from output inspection.

Exchangeability fails across driving sessions.. Conformal validity is conditional
45 on exchangeability between calibration and test data. Real driving data arrives in sessions — drives — that differ persistently in route, traffic, lighting and weather. On CADC [1] we measure per-drive miscoverage under a single pooled quantile ranging from 0.083 to 0.623 against a target of 0.20: a $7.5\times$ spread across drives that are all nominal, all clear-road, several sharing a collection
50 date (Section 4.3). Between-session variation is as large as the weather effect the monitor exists to detect. The guarantee stays formally correct — it holds on average over the calibration distribution — and becomes operationally useless, because the monitored quantity is dominated by which session the vehicle is on.

For navigation specifically this is the worst way for a monitor to fail. An
55 alarm that tracks route identity rather than perception loss will fire on an unremarkable drive and stay silent on a dangerous one, and a planner wired to react to it inherits that error directly. Worse, a monitor with undemonstrated sensitivity that is nonetheless counted in a safety argument is more dangerous than no monitor at all, because the false assurance displaces other mitigations.

60 *What we can and cannot say about the weather effect..* Our original objective requires a detectable difference between nominal and degraded conditions. On drive-disjoint held-out data, with resampling clustered over both nominal sessions and degraded drives, we measure $+0.064$ with a 95% interval of $[-0.061, +0.193]$. This is an *underpowered null*, not a demonstrated absence, 65 and we are careful to report it as such throughout: the interval is consistent with a real effect of the size we previously claimed and equally consistent with none. Nine evaluation sessions cannot resolve an effect of this magnitude — the minimum detectable effect is roughly twice the point estimate.

We also report why an earlier version of this measurement appeared significant. 70 Reporting the interval both with and without the degraded arm resampled shows that the earlier significance came from two compounding estimator defects rather than from a larger effect (Section 4.5). We suggest that dual reporting as a general practice for clustered effect estimates: it is what allows a reader to distinguish an underpowered design from a genuine absence, which the point 75 estimate alone never does.

Scope and honesty about this work’s own history.. Both findings above were discovered by controls applied to our own pipeline, and the numbers those controls invalidated were ours. We think this strengthens rather than weakens the case for the controls, but it puts an obligation on us to be exact about what is being 80 corrected: the affected results were obtained in an unpublished predecessor of this work and were never published, so no external record requires correction. Snowy Scenes [2] appears throughout as a negative control — its real detector shows no weather effect while its null detector shows a large one, the mirror image of CADC — and that contrast is what shows the diagnostic discriminates 85 between datasets rather than reporting a property of our own harness.

2. Related Work

2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature concatenation toward BEV-centric architectures that project both modalities into
90 a shared spatial representation before fusing, with attention-based mechanisms used to weight each modality’s contribution per region. A separate line of work has shown that such fusion is not automatically robust: a corrupted or adversarially perturbed single modality can dominate a static or softly-attended fusion decision and collapse detection performance, since nothing in the fusion mechanism
95 verifies that the modalities being combined actually agree geometrically before combining them. The detector this paper monitors adopts a verify-then-fuse design intended to address exactly that failure mode – a Cross-modal Consistency Probe that computes a per-BEV-cell consistency score between camera and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines
100 a disagreeing modality at the grid-cell level, trained under an Adversarial Consistency Training objective. We treat this design here strictly as background infrastructure and describe only as much of its mechanism (Section 3) as this paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic.

2.2. Conformal Prediction and Anytime-Valid Sequential Testing

Split conformal prediction [3, 4] gives distribution-free, finite-sample coverage guarantees for a black-box predictor’s outputs by calibrating a nonconformity score on held-out exchangeable data; the guarantee is a batch one, valid at a single fixed evaluation, and does not by itself say anything about monitoring
110 a live, continuously arriving stream. Sequential testing by betting [5, 6] closes that gap: framing a hypothesis test as an accumulating wealth process that is a nonnegative supermartingale under the null lets Ville’s inequality bound the false-alarm probability uniformly over all stopping times, giving an anytime-valid guarantee with no fixed monitoring horizon and no post-hoc multiple-testing correction. The direct predecessor for applying this backbone to a live
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perception-failure problem is Monroy Muñoz, Verma, and Timans’s work on sequential object-tracking-failure detection [7], which frames tracking failure as an e-process over a bounded tracking-quality metric with the betting rate learned adaptively from the metric’s own history (approximate GRAPA or Scale-Free Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks. It is a strong, directly applicable baseline, and everywhere in this paper that we compare against a "covariate-blind" betting rule, we mean their rule, reimplemented exactly, not a strawman.

Adjacent recent work conditions conformal guarantees on covariates or system state without adopting the sequential/anytime-valid framing: state-dependent conformal prediction for perception-based autonomous systems [8] combines conformal error bounds with symbolic reachability analysis but remains a batch procedure; context-aware nonconformity functions for robotic planning and perception [9] learn context-conditioned nonconformity scores, with context drawn from learned task representations rather than a cross-modal sensor-disagreement signal specifically; and a value-gated modality-refiner approach [10] uses cross-modal agreement to gate fusion, but as a learned neural gate rather than a statistical test covariate, and targets sentiment/action-recognition/audio-visual tasks rather than autonomous-vehicle perception. None of these combine sequential anytime-valid testing, spatially-resolved multi-object structure with multiplicity control, and a betting rule conditioned on an external cross-modal disagreement signal – the combination this paper’s monitoring layer targets, following the sharpened contribution statement worked out in the ICRA-targeted predecessor to this paper’s monitoring contribution.

2.3. Conditional Validity and Negative Controls

Marginal conformal validity holds on average over the calibration distribution and can fail badly within identifiable subgroups. Mondrian conformal prediction [11] addresses this by conditioning calibration on a taxonomy, giving validity within each category rather than marginally; the textbook treatment appears in Vovk, Gammerman and Shafer [3]. The taxonomy is usually a class

label. We apply it with the driving session as the category, which is the grouping along which we measure exchangeability to fail in real autonomous-driving data (Section 4.3), and we report a case where the resulting benefit reverses.

We are not aware of prior work that reports a null-detector control for a conformal perception monitor – that is, rerunning the pipeline with the monitored model’s predictions replaced by a constant to establish that the monitor responds to model error rather than to dataset structure. The closest established practice is the use of permutation and label-shuffling baselines in the broader evaluation literature, which serve the same epistemic purpose: distinguishing a method’s measured effect from structure already present in the data. Our contribution here is not the idea of a negative control but the specific, cheap instantiation for conformal monitoring, together with evidence that it changes conclusions on real published-style numbers (Section 4.2).

2.4. Adverse-Weather Datasets for Autonomous Perception

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and Waymo Open Dataset predominantly capture clear-weather conditions, limiting their use for evaluating perception under real distribution shift from adverse weather. Two existing datasets focus specifically on snowy conditions: the Canadian Adverse Driving Conditions dataset (CADC) [1], the first fully snowy dataset built for 3D object detection, and the Winter Adverse Driving dataSet (WADS), the first to add 3D semantic segmentation and individual-snowflake labeling; both, however, are reported to have comparatively low-resolution or unimodal sensing relative to more recent captures. Bijelic et al.’s fog- and rain-focused multimodal fusion work [12] addresses an adjacent adverse-weather regime with a different sensing challenge (fog/rain rather than snow) and is not built around a clear-to-degraded onset transition. Snowy Scenes [2], described fully in Section 4, is a substantially more recent (2026) multimodal dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB, and thermal cameras; the price of that specificity, as we found once real access arrived, is that it contains no clear-weather split at all, which this paper ad-

addresses by measuring a real, physically grounded severity ordering across its own weather categories rather than assuming the dataset would hand us the clear/snow contrast originally planned around. This paper’s own contribution to that evaluation landscape is not a new dataset. We apply the null-detector
180 diagnostic to both Snowy Scenes and CADC and it returns opposite verdicts on them, which is what shows it discriminates between datasets; the monitoring measurements themselves rest on CADC alone, and Snowy Scenes serves as a negative control rather than a second positive result.

3. Method

185 3.1. Background: The Monitored Detector

The detector this paper monitors is a camera/LiDAR BEV fusion architecture; we summarize only the components this paper’s monitoring layer directly depends on, since the monitor itself is detector-agnostic and this paper’s own contribution begins with the monitoring layer described in the rest of this section.
190 tion.

Camera and LiDAR streams are each encoded into a shared bird’s-eye-view (BEV) feature grid, $F_{\text{cam}}, F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. A Cross-modal Consistency Probe (CCP) computes, per BEV grid cell (i, j) , a consistency score

$$S(i, j) = \sigma\left(\text{MLP}_S[F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|]\right) \in [0, 1], \quad (1)$$

from the concatenated ($\oplus$) camera and LiDAR feature vectors at that cell and
195 their absolute difference, so that $1 - S(i, j)$ is a cross-modal disagreement signal intended to flag localized geometric or semantic mismatch between the two modalities. A Gated Adaptive Fusion Module (GAFM) uses S to weight each modality’s contribution to the fused BEV representation at the grid-cell level, and an Adversarial Consistency Training (ACT) objective trains the detector,
200 the CCP, and the fusion gate jointly under a loss combining detection accuracy, a CCP contrastive-consistency term, and an attribution-regularization term, with adversarially perturbed camera and LiDAR branches included in the training

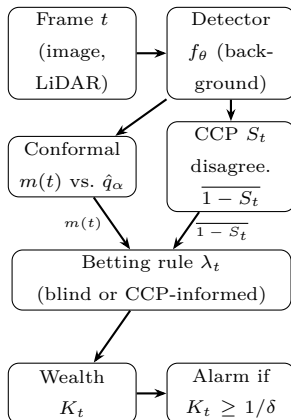


Figure 1: Per-frame monitoring pipeline. The detector f_θ (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score S_t ; everything downstream of it is this paper’s own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

loop to encourage S to remain informative under perturbation. This paper’s own contribution begins with the next subsection: everything from here on –
205 the conformal calibration, the sequential test, the spatial multiplicity control, and the use (and, as Section 4 reports, the eventual disuse) of S as a monitoring covariate – is this paper’s method, evaluated against real data in Section 4.

Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

210 3.2. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \dots$, one per LiDAR sweep. At every t , the base 3D detector f_θ described above produces per-object box predictions from the frame’s fused camera/LiDAR BEV features; the monitor itself is detector-agnostic and does not depend on f_θ ’s specific ar-
215 chitecture. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object i in frame t , the nonconformity score is the

ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually occupies,

$$s_i(t) = \|\hat{B}_i(t) - B_i(t)\|_1,$$

220 and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set's non-conformity scores $\{s_i\}_{i=1}^n$, so that a new object is covered iff $s_i(t) \leq \hat{q}_\alpha$. The frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty
225 frame carries no evidence of miscoverage. The null hypothesis the monitor tests is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ for all t ; the goal is to detect the first violation – weather-onset degradation – with an anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

230 3.3. The e -Process and Ville's Inequality

The wealth process is

$$K_0 = 1, \\ K_t = \prod_{s=1}^t (1 + \lambda_s(m(s) - \alpha)), \quad \lambda_s \in [0, \tfrac{1}{\alpha}] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably.} \quad (2)$$

We work through why this construction gives an anytime-valid test, rather than only stating the result, since the two steps – the supermartingale property and Ville's inequality – are each short but easy to state imprecisely.

235 *Step 1: K_t is a nonnegative supermartingale under H_0 .* Non-negativity requires $1 + \lambda_t(m(t) - \alpha) \geq 0$ for every realizable $m(t) \in [0, 1]$. Since $\lambda_t \geq 0$, the expression is affine and non-decreasing in $m(t)$, so its minimum over $m(t) \in [0, 1]$ occurs at $m(t) = 0$, giving $1 - \lambda_t \alpha$. This is non-negative exactly when $\lambda_t \leq 1/\alpha$ – the bound our implementation enforces on every betting

240 rule (`WealthProcess.lambda_max` in `conformal_monitor/betting.py`), and
the reason the bound is $1/\alpha$ specifically, not the also-plausible-looking $1/(1-\alpha)$
(which only bounds the opposite endpoint, $m(t) = 1$, and is too loose at the
endpoint that actually binds). For the supermartingale property itself, λ_t is cho-
sen $\mathcal{F}_{t-1}$ -measurably – it may depend on the entire history up to and including
245 frame $t - 1$, but not on $m(t)$ itself – so, conditioning on $\mathcal{F}_{t-1}$,

$$\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] = K_{t-1} \cdot \mathbb{E}[1 + \lambda_t(m(t) - \alpha) \mid \mathcal{F}_{t-1}] = K_{t-1}(1 + \lambda_t(\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] - \alpha)),$$

using that λ_t and K_{t-1} are both $\mathcal{F}_{t-1}$ -measurable and can be pulled outside
the conditional expectation. Under H_0 , $\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$, so the bracketed
term is ≤ 1 whenever $\lambda_t \geq 0$ (which it always is, by construction), giving
 $\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] \leq K_{t-1}$: precisely the supermartingale inequality, holding for *every*
250 choice of predictable, non-negative $\lambda_t \leq 1/\alpha$, which is exactly the freedom the
different betting rules in Sections 3.4–3.5 exploit without ever needing to reprove
this step.

Step 2: Ville’s inequality bounds the running maximum.. For a nonnegative
supermartingale K_t with $K_0 = 1$, Ville’s maximal inequality gives, for any
255 $c > 0$,

$$\mathbb{P}_{H_0}\left(\sup_{t \geq 0} K_t \geq c\right) \leq \frac{\mathbb{E}[K_0]}{c} = \frac{1}{c}.$$

Setting $c = 1/\delta$ gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$. The alarm rule – reject H_0 the
first time $K_t \geq 1/\delta$ – therefore has false-alarm probability at most δ *regardless*
of the (random) stopping time at which the process is examined: the guarantee
is over the entire trajectory’s supremum, not over K_t at one pre-specified t ,
260 which is exactly what makes checking continuously, with no fixed monitoring
horizon and no correction for how many times the test has been evaluated,
statistically valid rather than an uncorrected multiple-testing error. This two-
step argument – the supermartingale property from Step 1, Ville’s inequality
from Step 2 – is the established backbone of testing by betting and is not this
265 paper’s contribution; what varies across the rest of this section, and what this

paper’s own contribution actually depends on, is how the predictable sequence λ_t is chosen.

3.3.1. A Worked Numeric Example

To build intuition before the general betting rules, we trace K_t over five
 270 synthetic frames by hand, with $\alpha = 0.2$ (so $\lambda_{\max} = 5$) and a fixed, non-adaptive
 $\lambda_t = 2$ for simplicity (an actual bettor updates λ_t per Sections 3.4–3.5; holding
 it fixed here isolates what the recursion itself does). Suppose the observed
 miscoverage sequence is $m = (0.1, 0.15, 0.6, 0.55, 0.5)$ – nominal for the first two
 frames, then abruptly degraded from frame 3 onward, mimicking a weather
 275 onset at $t = 3$:

$$\begin{aligned} K_1 &= K_0(1 + 2(0.10 - 0.2)) = 1 \times 0.80 = 0.80, \\ K_2 &= K_1(1 + 2(0.15 - 0.2)) = 0.80 \times 0.90 = 0.72, \\ K_3 &= K_2(1 + 2(0.60 - 0.2)) = 0.72 \times 1.80 = 1.296, \\ K_4 &= K_3(1 + 2(0.55 - 0.2)) = 1.296 \times 1.70 = 2.203, \\ K_5 &= K_4(1 + 2(0.50 - 0.2)) = 2.203 \times 1.60 = 3.525. \end{aligned}$$

Two things are worth noticing in this trace. First, wealth *shrinks* while $m(t) < \alpha$
 (frames 1–2): a bettor staking $\lambda_t > 0$ on “miscoverage will exceed α ” loses
 money, by construction, when the pipeline is behaving nominally – this is the
 price of testing at all, and is what keeps the process a supermartingale rather
 280 than an unconditionally growing one. Second, wealth grows multiplicatively, not
 additively, once $m(t) > \alpha$ consistently (frames 3–5): each factor > 1 compounds
 against the previous wealth, which is why a real onset that persists for several
 frames can cross a large threshold $1/\delta$ (e.g. $1/\delta = 20$ for $\delta = 0.05$) well before
 $t \rightarrow \infty$, even though no single frame’s factor is large by itself. In this toy trace,
 285 $K_5 = 3.525$ would already trigger an alarm at $\delta = 0.3$ ($1/\delta \approx 3.33$) but not yet
 at $\delta = 0.1$ or $\delta = 0.05$ – illustrating concretely why detection delay lengthens as
 δ shrinks: a smaller false-alarm budget requires more accumulated evidence (a
 higher wealth threshold) before the anytime-valid guarantee permits an alarm.

3.4. Covariate-Blind Baselines

Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ itself, reproducing [7] as the baseline to beat. Approximate GRAPA, with $X_t = m(t) - \alpha$, sets

$$\lambda_t = \text{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X_{<t}^2] + \varepsilon}, 0, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on the log-wealth gradient, with $g_t = X_t/(1 + \lambda_t X_t)$ and $\eta_t = \lambda_{\max}/\sqrt{1 + \sum_{s \leq t} g_s^2}$,

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_t g_t, 0, \lambda_{\max}).$$

Both are exact reimplementations of the adaptive betting rules [7] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

3.5. Covariate-Informed Betting

Alongside the two contributions in Sections 3.6 and 4, we also investigated conditioning the betting rate on an external signal available at the same timestep as $m(t)$ but computed from independent evidence: the disagreement output of the monitored detector’s own CCP, $1 - S$ (Equation 1). We describe the mechanism because it motivates this paper’s monitoring design, and note here that it is specified and implemented but not evaluated in this paper (Section 4.7). Our covariate-informed bettor wraps either covariate-blind base bettor and scales its stake by the frame’s mean disagreement,

$$\lambda_t = \text{clip}\left(\lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)}), 0, \lambda_{\max}\right),$$

where $\kappa \geq 0$ gates the covariate’s influence ($\kappa = 0$ recovers the base bettor exactly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as

the lagging one the covariate-blind rules are stuck reacting to. Whether that intended mechanism holds on real data is not settled in this paper; the rule is
 315 specified and implemented, and no result for it is reported here (Section 4.7).

3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone wrong" – not useful to a downstream planner that needs to know *which* region of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$
 320 layout of cell-clusters (average-pooling per-cell miscoverage and disagreement down to this resolution) and run one independent sequential tester per cluster, each with its own wealth process and its own instance of whichever bettor is under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires multiplicity control or the overall false-alarm rate inflates with n ; we support
 325 two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell alarms, marks it permanently flagged – valid under arbitrary dependence but conservative, and the sticky-once-alarmed semantics is what we use for the per-cell detection-delay evaluation regardless of which regime produced the alarm. The e-BH alternative treats each cell's current wealth $K_t^{(i)}$ as an e-value: for
 330 a nonnegative supermartingale started at $K_0 = 1$, $\mathbb{E}_{H_0}[K_t^{(i)}] \leq \mathbb{E}_{H_0}[K_0^{(i)}] = 1$ at any fixed t (immediate from Section 3.3's Step 1, since the supermartingale inequality applied recursively from $t = 0$ gives $\mathbb{E}[K_t] \leq \mathbb{E}[K_{t-1}] \leq \dots \leq K_0 = 1$) – exactly the definition of an e-value under $H_0^{(i)}$: a nonnegative random variable with expectation at most 1. Given $n = n_h n_w$ such e-values at a fixed t , one per
 335 cell, e-BH [13] rejects the set

$$R_t = \left\{ i : K_t^{(i)} \geq \frac{n}{k^* q} \right\}, \quad k^* = \max \left\{ k \in \{0, \dots, n\} : e_{(k)} \geq \frac{n}{k q} \right\},$$

where $e_{(1)} \geq e_{(2)} \geq \dots \geq e_{(n)}$ is the descending order statistic of the n current wealths, and $k^* = 0$ gives $R_t = \emptyset$. The rule is a direct e-value analogue of Benjamini-Hochberg: it thresholds the k -th largest e-value against a level that grows with k (more permissively for larger candidate rejection sets), then
 340 commits to the *largest* such k – the same "largest crossing index" structure as

classical BH’s p -value threshold, but with the inequality direction reversed because e -values are large, not small, under evidence against the null. Its false discovery rate guarantee, $\mathbb{E}[\text{FDR}_t] \leq q$ at every fixed t , holds under *arbitrary dependence* among the n e -values – unlike classical BH on p -values, which requires independence or specific positive-dependence structures (PRDS) to control FDR without an extra $\log n$ correction factor – because the e -value definition ($\mathbb{E}[e_i] \leq 1$ under $H_0^{(i)}$) is closed under averaging regardless of the joint dependence structure of the e_i , which is precisely the guarantee needed here: spatially adjacent BEV cells’ miscoverage and disagreement signals are driven by overlapping physical causes (a single snow squall affects many neighboring cells at once) and are certainly not independent. Unlike Bonferroni’s sticky once-alarmed flags, e -BH is applied fresh at every timestep from each cell’s current wealth, so it produces a live, non-sticky map that can un-flag a cell once its local evidence subsides – appropriate for a spatial map meant to reflect *current*, not historical, trustworthiness. Algorithm 1 summarizes the per-frame update.

3.6.1. Betting-Rule Comparison: Theoretical Properties and Tradeoffs

Table 1 summarizes the three betting rules evaluated in this paper, all valid choices of the predictable λ_t in Equation 2 (so all three inherit Section 3.3’s anytime-valid guarantee unconditionally; what differs is statistical power, not validity). AGRAPA is a plug-in estimator of the (asymptotically) growth-rate-optimal constant bet, tracking the running first and second moments of $X_t = m(t) - \alpha$ and converging quickly when the true miscoverage rate is close to stationary, but reacting relatively slowly to an abrupt regime change, since a single new observation only nudges a running average computed over all prior history. SF-OGD instead takes an online-convex-optimization view, performing AdaGrad-style gradient ascent directly on (an approximation to) the log-wealth objective with a decaying step size; this makes it more reactive to recent observations by construction (the step size η_t shrinks as $1/\sqrt{t}$, but each step still moves λ_t in the direction the *most recent* gradient indicates), at the cost of higher variance in λ_t itself, particularly early in a stream before $\sum_{s \leq t} g_s^2$ has accumulated

Table 1: Betting rules compared in this paper. All three are valid choices of λ_t in Equation 2 and inherit the same anytime-valid guarantee unconditionally (Section 3.3); they differ in statistical power, not validity.

Rule	Mechanism	Reactivity / stability
AGRAPA	running 1st/2nd moment plug-in	smooth, slower to react to abrupt shift
SF-OGD	AdaGrad-style gradient ascent on log-wealth	more reactive, higher early-stream variance
CCP-informed	multiplicative covariate gain κ on either base rule	inherits base rule's tradeoff; adds covariate sensitivity (specified and implemented; not evaluated here)

enough to stabilize η_t . Both are exact reimplementations of the rules [7] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on. The CCP-informed variant wraps either base rule multiplicatively rather than replacing it, so it inherits
375 whichever base rule's reactivity/stability tradeoff was chosen, and adds a single scalar gain κ controlling how strongly the covariate perturbs the base bet; $\kappa = 0$ recovers the base rule exactly, which would make a covariate-informed versus covariate-blind comparison a controlled one (same base rule, same wealth-process mechanics, differing only in whether $\overline{1 - S_t}$ enters the stake) rather than
380 two unrelated procedures, were such a comparison reported.

Algorithm 1 Per-frame spatial monitoring update

Require: frame t ; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w) ; correction mode

- 1: compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
- 2: compute per-cell CCP disagreement $\overline{(1 - S)}^{(i)}(t)$
- 3: **for** each cluster $i = 1, \dots, n_h n_w$ **do**
- 4: $\lambda_t^{(i)} \leftarrow \text{bettor.NEXT_LAMBDA}(\overline{(1 - S)}^{(i)}(t))$
- 5: $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot (1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha))$
- 6: $\text{bettor.UPDATE}(m^{(i)}(t), \overline{(1 - S)}^{(i)}(t))$
- 7: **end for**
- 8: **if** correction = Bonferroni **then**
- 9: flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
- 10: **else** ▷ e-BH
- 11: flag the top- k^* cells by $K_t^{(i)}$, $k^* = \text{largest } k \text{ with } K_{(k)} \geq (n_h n_w) / (k \delta)$
- 12: **end if**
- 13: **return** flagged-cell map (untrustworthy regions at time t)

3.7. Session-Conditional (Mondrian) Calibration

Split conformal calibration pools nonconformity scores and produces one quantile, valid on average over the calibration distribution. Real driving data arrives in sessions that differ persistently in weather, lighting, traffic and route, and Section 4.3 measures the resulting per-session violation directly. Mondrian conformal prediction [11, 3] conditions calibration on a taxonomy, giving validity within each category; here the category is the driving session.

Mondrian requires calibration data in every category it is queried about, which forces a within-session split: each session calibrates on its own leading frames and is monitored on the remainder. This is the deployment-faithful arrangement — a vehicle calibrates on the early known-nominal portion of the drive it is currently on, rather than assuming a quantile estimated on previous drives transfers — and degraded frames are scored against the quantile of the session the stream began in, which is what a deployment does when conditions change mid-drive. Because calibration and monitored frames then sit adjacent in time within a correlated stream, we insert a temporal gap between them and sweep its size; an effect inflated by adjacency shrinks as the gap grows.

3.8. A Phantom-Detection Nonconformity Score

The score of Section 3.2 is a box-regression residual restricted to cells a
400 ground-truth object occupies, and is therefore structurally blind to spurious
detections in empty space. Snow’s dominant effect on LiDAR is precisely spu-
rious near-range returns, so we add a second score family: the predicted peak
class activation at cells containing no ground-truth object, calibrated by the
same split-conformal procedure at level α . The frame-level rate is the convex
405 combination

$$m(t) = (1 - w) m_{\text{loc}}(t) + w m_{\text{phantom}}(t), \quad (3)$$

with $w \in [0, 1]$. Each component is calibrated at level α and so has expectation
at most α under the nominal distribution; a convex combination of two such
quantities inherits the bound, which is exactly the H_0 condition the wealth pro-
cess of Section 3.3 requires. No new theory is needed: the conformal guarantee
410 is agnostic to the choice of score function. Setting $w = 0$ recovers the original
score exactly.

3.9. Block Aggregation

Averaging m over blocks of k consecutive frames before the betting step
trades temporal resolution for reduced variance in the quantity the bettor con-
415 sumes. Under independent frames the reduction would be $\sqrt{k}$; driving frames
are strongly correlated, so the realised gain is smaller; we do not report a mea-
surement of it here. We keep one trajectory entry per frame regardless of k , so
detection delays remain in frame units and are comparable across block sizes
and against the unblocked baseline.

4. Experiments

Every number below is produced by a script committed alongside the manuscript
and written to a JSON artifact named in the corresponding caption.

4.1. Setup

CADC [1] provides 32 downloaded driving sessions across three collection
425 dates: 18 nominal (**bare**-road) and 14 snow-covered. Snowy Scenes [2] provides real snowy driving; it contains no clear-weather split, so its contrast is constructed across weather categories and it serves here as a negative control rather than a second positive result.

Train/evaluation separation.. The detector and the monitor are built on disjoint
430 *drives*. Frame-level holdout would not suffice: consecutive CADC frames are ~ 10 Hz samples of one scene, so a frame-level split leaves near-duplicates of training frames in evaluation. Drives are interleaved within each weather category, giving 16 training and 16 evaluation drives (1534 / 1408 frames) with both categories and all three dates represented on each side. Calibration and
435 monitoring both draw only from evaluation drives, so the conformal scores are out-of-sample.

This matters because an earlier version of this work did not do it. The detector and the monitor were constructed over one unsplit dataset object, so every calibration, nominal and degraded frame had been a training frame, and the
440 nonconformity scores were in-sample residuals against which a calibrated quantile bounds nothing. A regression test now asserts that the train and evaluation frame-id sets are disjoint.

Nesting with session-conditional calibration.. Mondrian calibration needs calibration frames from inside each monitored session, so an evaluation drive is
445 split again internally into a leading calibration prefix, a temporal gap, and a monitored remainder. That inner split lives entirely within evaluation drives and never reaches training data.

Inference.. Sessions, not frames, are the unit of analysis: frames within a drive are strongly correlated and frame-level resampling understates intervals badly.
450 Where an estimate contrasts nominal against degraded data, *both* arms are resampled — nominal sessions and degraded drives — because the degraded

Table 2: Null-detector diagnostic. The nominal-to-degraded jump in m is identical to 17 significant figures whether the box head’s real output or zeros are used, on both datasets — the monitor’s evidence does not depend on the detector at all. Artifacts: `detector_collapse_cadc.json`, `detector_collapse_snowy.json`.

Dataset	real detector	zeroed detector	difference
CADC	0.13904668284371138	0.13904668284371138	0
Snowy Scenes	−0.15573322510822504	−0.15573322510822504	0

side carries real between-drive variation (measured sd 0.057) that a fixed pool would silently treat as zero. All intervals are percentile bootstrap.

4.2. The Null-Detector Diagnostic

455 The diagnostic reruns the entire pipeline, calibration included, with the detector’s box-regression output replaced by zeros. The nonconformity score then reduces to $\|B_{\text{target}}\|_1$ and depends only on ground-truth scene content. If a monitor’s behaviour reflects detector error, zeroing the detector must change it.

Table 2 gives the result on the checkpoints that produced our earlier numbers. The agreement is exact, not approximate: the detector contributes nothing 460 measurable to the monitor’s evidence, because those checkpoints had collapsed under a placeholder detection loss and *were* null detectors. This is the cleanest available statement of the finding, and it is immune to any question about evaluation-harness configuration, since it compares two branches of a single deterministic computation. 465

We draw a conclusion beyond our own case. A conformal monitor built on a degenerate score does not fail conspicuously — it reports calibrated quantiles and plausible alarm times throughout. One extra evaluation pass distinguishes the two situations unambiguously. We suggest it be reported as standard, with one scope condition stated honestly: this is a null for scores built from regression 470 residuals. A score reading a different output head needs its own null, and Section 5.4 explains why constructing one is not always straightforward.

Table 3: Per-drive mean frame miscoverage under one pooled quantile ($\hat{q} = 1.935$, target $\alpha = 0.20$). All **bare** drives are nominal clear-road driving. Artifact: `per_drive_miscoverage_cadc.json`.

Group	n drives	mean m (sd)	range
calibration drives	9	0.252 (0.140)	0.133 – 0.623
nominal drives	9	0.181 (0.061)	0.083 – 0.284
degraded drives	14	0.238 (0.057)	0.128 – 0.329

4.3. Cross-Session Exchangeability Failure

Marginal validity holds on average while failing per session. Four of nine
475 nominal drives sit above α ; one calibration drive reaches 0.623 against the quantile it helped define; the spread across all **bare** drives is $7.5\times$. The comparison that matters is against the effect being sought: the degraded-minus-nominal difference is $+0.057$ while the between-drive standard deviation is 0.061, and six of nine nominal drives fall inside the degraded drives’ range. A monitor calibrated
480 marginally across sessions is therefore substantially detecting session identity, and its alarms carry route information rather than perception information.

4.4. Session-Conditional Calibration, and Where It Reverses

Mondrian conformal prediction [11] conditions the quantile on a taxonomy; here the taxonomy is the driving session.

On the 5-epoch checkpoint session-conditional calibration more than halves
485 the worst-case per-session deviation. On the converged checkpoint the ordering reverses and Mondrian is roughly $1.7\times$ worse. We do not have an explanation and do not offer one. Two consequences follow, and we state both rather than the convenient one. First, the benefit is a property of the model–data combination, not an unconditional improvement: any claim that Mondrian improves
490 worst-case per-session coverage is contradicted by our own second measurement. Second, neither the benefit nor the reversal carries an interval — $\max_s |m_s - \alpha|$ is a maximum over nine sessions, among the least stable estimators available —

Table 4: Mondrian versus marginal calibration, matched on sessions, frames, gap and cached scores, differing only in which quantile scores each session. The advantage on one checkpoint reverses on the other. Artifacts: `mondrian_gap_sweep_cadc.json`, `mondrian_gap_sweep_cadc_converged.json`.

gap	effect / nominal sd		worst $ m - \alpha $	
	marginal	Mondrian	marginal	Mondrian
<i>5-epoch checkpoint</i>				
0	0.551	0.849	0.412	0.187
10	0.528	0.848	0.427	0.190
20	0.558	0.829	0.480	0.200
<i>converged checkpoint</i>				
0	-0.033	-0.025	0.187	0.344
10	-0.104	-0.079	0.221	0.363
20	0.096	0.003	0.210	0.353

so the reversal may itself be noise. A practitioner should measure this on their
 495 own model rather than assume either direction.

4.5. The Weather Effect: an Underpowered Null

The effect is not distinguishable from zero at any gap, under either calibration scheme, and the null detector is likewise null throughout.

This is an underpowered null, not a demonstrated absence.. The interval $[-0.061, +0.193]$
 500 is consistent with a real effect of the size we previously claimed and equally consistent with none. With nine evaluation sessions and a between-session sd near 0.06, the minimum detectable effect at 80% power is roughly 0.13 — about twice the point estimate. The design cannot resolve an effect of this magnitude, and no conclusion about the presence or absence of weather-induced degradation
 505 should be drawn from it.

Why an earlier version appeared significant.. Before correction this measurement read +0.074 with an interval of $[+0.020, +0.117]$. The point estimate

Table 5: Nominal-versus-degraded effect on drive-disjoint held-out drives (9 nominal sessions, 7 degraded drives), Mondrian calibration. The two-level interval resamples nominal sessions and degraded drives; the nominal-only interval resamples sessions alone, from the same draws, and is shown so the effect of the correction is visible. Artifact: `split_effect_split_ep5.json`.

gap	effect	two-level 95% CI	nominal-only 95% CI	inflation	null detector
0	+0.064	$[-0.061, +0.193]$	$[-0.031, +0.143]$	$1.46\times$	n.s.
10	+0.076	$[-0.059, +0.211]$	$[-0.029, +0.161]$	$1.43\times$	n.s.
20	+0.069	$[-0.073, +0.210]$	$[-0.039, +0.163]$	$1.40\times$	n.s.

barely moved — $+0.074 \rightarrow +0.064$, a 14% reduction — so removing the train/evaluation overlap did *not* destroy the effect. Significance was lost for two separate statistical reasons, and reporting both intervals separates them. Restricting to held-out drives cut the session count from 18 to 9, which alone widened the interval enough to cross zero (the nominal-only column). Resampling the degraded arm widened it a further $1.4\text{--}1.5\times$. Neither correction alone would have exposed the problem, and the earlier significance was an artifact of the two compounding.

We suggest reporting both intervals as general practice for clustered effect estimates. A single interval tells a reader whether a result is significant; the pair tells them *why* it is not, which is the difference between concluding “underpowered” and concluding “absent.”

4.6. Snowy Scenes as Negative Control

On Snowy Scenes the real detector’s effect is -0.0085 with a 95% interval of $[-0.0719, +0.0553]$, centred on zero, while the null detector’s is -0.1315 , $[-0.1917, -0.0726]$ (`effect_exists_snowy.json`). The only signal present is a content artifact pointing the wrong way, which is why the null monitor “detected” there while the real one did not.

This is the mirror image of CAD/C, where the real effect is the larger and the null’s interval covers zero. The same diagnostic returns opposite verdicts on two datasets, which is the evidence that it discriminates between them rather

than reporting a property of our pipeline. We note that these two intervals
 530 were computed with frame-level rather than session-level resampling and are
 therefore narrower than our convention elsewhere would give; the qualitative
 contrast is robust to this, but the Snowy interval widths should not be read as
 comparable to Table 5’s.

4.7. *What We Do Not Report*

535 Three lines of work were carried far enough to have numbers and are ex-
 cluded, because the numbers do not support conclusions.

A curve relating measured weather effect to detector quality across check-
 points is omitted: its quality axis (heatmap object-versus-empty separation)
 shares a term with one of the two scores plotted against it — the phantom score
 540 *is* the empty-cell activation that the axis subtracts — so a decline along that
 axis is partly circular. Its inputs also predate the corrections in Section 4.1. We
 record it as an open question, not a finding.

A covariate-informed betting rule conditioned on the detector’s own cross-
 modal consistency score, and a spatially resolved variant using per-cell e-processes
 545 under multiplicity control, are described in Section 3 because they motivate the
 design, but neither is evaluated here and neither carries a result in this paper.

Operating-curve and detection-delay results are omitted because, under the
 corrected design, no configuration has a demonstrable false-alarm advantage:
 with 36 replicates, an observed rate of 0/36 has an exact upper confidence
 550 bound of 0.080, which does not establish control at $\delta = 0.05$.

5. Discussion

5.1. *Why a Fictitious Monitor Looks Healthy*

The mechanism is worth stating precisely, because it generalizes past our
 particular bug. The nonconformity score is a residual evaluated where a ground-
 555 truth object sits. When the prediction term vanishes it becomes $\|B_{\text{target}}\|_1$: a
 function of the labels alone. Nothing downstream notices. Split conformal

calibration still returns a well-defined quantile, because it is the quantile of *some* exchangeable score distribution. The betting process still accumulates wealth, because the miscoverage rate still moves between the nominal and degraded
560 halves of a stream. The alarm still fires at a plausible time, because ground-truth box statistics genuinely differ between a dataset’s splits. Every component behaves exactly as specified, and the monitor reports annotation structure as perception degradation.

We were careful about which dataset property drives this, having initially
565 got it wrong. We had attributed it to object count roughly doubling between CADC’s nominal and degraded splits. Direct measurement contradicts that: in the scored bird’s-eye-view window, degraded frames contain *fewer* matched objects than nominal ones ($0.84\times$). Since the score reduces to $\|B_{\text{target}}\|_1$, the driver must instead be the distribution of ground-truth box dimensions and
570 positions across the two collection dates. That is a more interesting explanation than density, and it interacts directly with the date confound discussed below.

There is an inversion here that matters for how the failure is detected. On CADC the null monitor was *better behaved* than the real one — a zero false-alarm rate where the real monitor alarmed on everything. Scene content is relatively
575 homogeneous across nominal drives, so a content-derived score transfers cleanly between sessions; genuine detector error is heterogeneous across sessions and does not. The artifact was better behaved than the truth. This inverts the usual heuristic that cleaner numbers indicate a better method, and it is exactly why inspecting outputs cannot substitute for a control.

580 5.2. Sessions Are Not Exchangeable, and What It Costs Navigation

Conformal validity is conditional on exchangeability, and driving data violates it in a structured way: it arrives in sessions that differ persistently in route, traffic, lighting and weather. Our measurement puts a number on the cost — a $7.5\times$ per-drive spread in miscoverage under a pooled quantile, with
585 between-session variation as large as the weather effect being sought.

The consequence is not a modest loss of efficiency. It is that the monitored

quantity is dominated by which session the vehicle is on, so the monitor detects session identity. For a navigation stack this is the worst available failure mode. An alarm that tracks route rather than perception fires on an unremarkable drive and stays silent on a dangerous one, and a planner wired to react — by
590 slowing, handing back, or initiating a minimum-risk manoeuvre — inherits that error directly and acts on it. A monitor whose sensitivity has not been demonstrated but which is nonetheless counted in a safety argument is worse than no monitor, because the assurance it appears to provide displaces mitigations that
595 would otherwise be required.

Two practical consequences follow for anyone deploying this pattern. First, calibration should be conditioned on the session, or at minimum validated per session rather than in aggregate — an aggregate coverage check passes while individual sessions sit far from target. Second, because our own attempt at the
600 session-conditional fix reversed direction between two checkpoints of the same model (Section 4.4), the check has to be repeated against the model actually being shipped, not inherited from a paper.

5.3. *Underpowered Is Not Absent*

Our headline measurement is a null we decline to interpret as an absence, and
605 the distinction carries the section. An interval of $[-0.061, +0.193]$ around a point estimate of $+0.064$ excludes neither a real effect of the size we previously claimed nor no effect at all. With nine evaluation sessions the minimum detectable effect is roughly twice the estimate, so failing to detect it is what this design does regardless of whether the effect is there.

The dual-interval reporting in Table 5 is what makes that diagnosable rather
610 than assertable. Because the nominal-only and two-level intervals are computed from the same session draws and differ only in whether the degraded arm is resampled, a reader can attribute the loss of significance to its causes: the session count falling from 18 to 9 under the drive-disjoint split, and a further
615 $1.4\text{--}1.5\times$ widening from clustering the degraded arm. The point estimate moved by 14%. We suggest reporting both intervals whenever an effect is estimated

from clustered data on both sides; a single interval says whether a result is significant, the pair says why not.

It is worth being explicit that the correction which mattered most was not the one we expected. Removing the train/evaluation overlap — the most serious
620 defect, and the one that voids the earlier numbers as a matter of principle — barely moved the point estimate. The significance was carried by estimator defects, not by the leakage. Both had to be fixed; only one turned out to change the number.

625 5.4. Limitations

Scope of the null-detector diagnostic.. It is a null for scores built from regression residuals. A score reading a different output head requires its own, and constructing one is not always straightforward: for a score built from classification-head activation, zeroing that head makes every score identical and the statistic
630 degenerate, which is a floor rather than an informative control. We report this because it bounds the recommendation, and a general recipe — sever the dependence between the score and the model output while leaving the calibration pipeline intact — is easier to state than to instantiate for every score family.

The diagnostic does not catch every failure.. It catches a monitor that is not
635 reading the model. It does not catch a monitor that reads the model but measures the wrong thing — including the session-identity failure documented in Section 4.3, which passes the null-detector check.

Weather is confounded with collection date.. CADC’s degraded drives come from a different date than its nominal ones, so sensor condition, route and
640 traffic are confounded with weather. The null-detector control bounds the contribution of label content specifically; it does not bound the rest. The dataset publishes a quantitative per-drive snowfall-severity count that varies *within* date and would break this aliasing; we identified it too late to use, and regard it as the most promising next step.

645 *Scale and single-run training.* Nine evaluation sessions, one detector architecture, one dataset carrying the positive findings, and one training run — no seed replication, so nothing here has error bars in that dimension. Training used chunked resumption, so optimizer moment estimates re-warm at each resume, a small deviation from an uninterrupted run.

650 *Statistical scope.* The Snowy Scenes intervals in Section 4.6 use frame-level resampling and are narrower than our session-level convention would give. The Mondrian worst-case statistic is a maximum over nine sessions and carries no interval. We report both rather than presenting the numbers as comparable to the clustered estimates elsewhere.

655 **6. Conclusion**

We set out to reduce the detection delay of a conformal, anytime-valid monitor for camera/LiDAR 3D perception under real adverse weather, so that a navigation stack could be told sooner when its perception had stopped being trustworthy. We did not achieve that. What the attempt produced instead are
660 two findings about how such monitors fail silently, and a measurement we report as unresolved rather than settled.

First, a conformal monitor can produce calibrated quantiles and plausible alarm times while measuring nothing about the detector. When the prediction head degenerates, the nonconformity score becomes a function of the ground-truth labels alone and the monitor reports annotation structure as perception
665 degradation, with nothing in its output revealing this. The *null-detector diagnostic* — rerun the pipeline with the prediction head’s output replaced by zeros — exposes it for the cost of one evaluation pass, and on our own earlier results reproduced them bit-identically, to 17 significant figures, from a model
670 that outputs nothing. On one dataset the null monitor was better behaved than the real one, which is precisely why output inspection cannot substitute for the control.

Second, split conformal’s exchangeability assumption fails across driving sessions. Per-drive miscoverage under a pooled quantile spans $7.5\times$ on nominal clear-road drives, with between-session variation as large as the weather effect being sought. The guarantee remains formally correct and becomes operationally useless: the monitor detects which session the vehicle is on. For navigation that is the worst available failure mode, because a planner wired to a monitor that tracks route identity will act on route identity.

On the weather effect we report an underpowered null and are explicit that this is not an absence: $+0.064$ with a 95% interval of $[-0.061, +0.193]$ on drive-disjoint held-out data, consistent both with a real effect of the size we previously claimed and with none. We also show why an earlier version appeared significant, by reporting the interval with and without the degraded arm resampled — a decomposition we recommend for clustered estimates generally, because it distinguishes an underpowered design from a genuine absence where a single interval cannot.

Three directions follow, in the order we would pursue them. The most promising is already identified: CADC publishes a quantitative per-drive snowfall-severity count that varies within collection date, which would replace the date-aliased binary contrast used here with a dose-response regression — more efficient statistically, and free of the confound that limits every effect estimate in this paper. Second, the dataset releases substantially more drives than we downloaded, and since the number of evaluation sessions is the binding constraint on every interval reported here, raising it addresses the limitation more directly than any change of estimator. Third, the null-detector diagnostic should be generalized from a single check into a family: a null per score family, and ideally a graded sensitivity profile obtained by injecting known degradations of known magnitude, so that a null result acquires a measured detection floor rather than resting on an instrument of unknown sensitivity.

We would emphasise one point above the others. Every failure in this work was silent — a collapsed detector, an evaluation harness whose replicates were identical, a monitor reading annotation statistics, calibration valid in aggregate

and broken per session, and a train/evaluation overlap that made every confor-
705 mal score in-sample. In each case the outputs looked reasonable, and in each
case what exposed the problem was a control: zeroing the detector, asserting
that replicates differ, measuring per session rather than in aggregate, asserting
that train and evaluation frames are disjoint. For a monitoring system, whose
whole purpose is to be trusted when nothing else is watching, we think the con-
710 trols deserve as much space in the reporting as the headline numbers — and
that a monitor whose sensitivity has never been demonstrated should not be
counted in a safety argument at all.

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